Alberto Andrés Valdés González.
Degree: Mathematical Engineer.
Work position: ML-Engineer.
Mail: anvaldes@uc.cl/alberto.valdes.gonzalez.96@gmail.com
Location: Santiago, Chile.

LLM vs Diffusion Models

Setting: both are **generative models**: they learn a data distribution $p(x)$ and sample new data from it. They differ in *how* they generate: an LLM writes **one token at a time**, left to right; a diffusion model starts from **pure noise** and refines the **whole sample** step by step.

LLM (autoregressive): text is a sequence of discrete tokens $x_1, \dots, x_T$; with the chain rule, a Transformer learns to predict the next token from the previous ones (cross-entropy):

$$p_\theta(x) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}), \qquad \mathcal{L} = -\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t})$$

Generation: sample x_t, append it, repeat; T tokens need T sequential forward passes.

Diffusion: a **forward process** gradually adds Gaussian noise to a real sample x_0 (e.g. an image) over T steps until it is pure noise; a network ϵ_θ (U-Net or Diffusion Transformer) learns to **predict the noise** added, so it can undo it:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \qquad \mathcal{L} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|^2\right]$$

Generation: start with $x_T \sim \mathcal{N}(0, I)$ and denoise $x_T \to \cdots \to x_0$ (typically 20–50 steps), each step updating *all* pixels in parallel. The condition c (a text prompt embedding) steers the result (classifier-free guidance); most models work in a compressed **latent space** (Latent Diffusion).

	LLM (autoregressive)	**Diffusion**
Generates by	Next-token prediction, left to right	Iterative denoising from random noise
Data type	Discrete (tokens): text, code	Continuous: images, audio, video
Training loss	Cross-entropy on the next token	MSE on the predicted noise
Parallelism at inference	Sequential across tokens (T passes)	Parallel across the sample, sequential across denoising steps
Editing	Hard to revise earlier tokens	Easy: inpainting, image-to-image, refine regions
Strengths	Reasoning, instructions, dialogue, tool use	High-fidelity, diverse visual and audio samples
Examples	GPT, Claude, Llama, Gemini	Stable Diffusion, DALL·E, Imagen, video models

The boundary is blurring: diffusion models often use a **Transformer** backbone (DiT), **diffusion language models** denoise masked text tokens in parallel, and some image models are autoregressive. The core difference is the **generation procedure**, not the architecture: piece by piece vs sculpting the whole output from noise.